

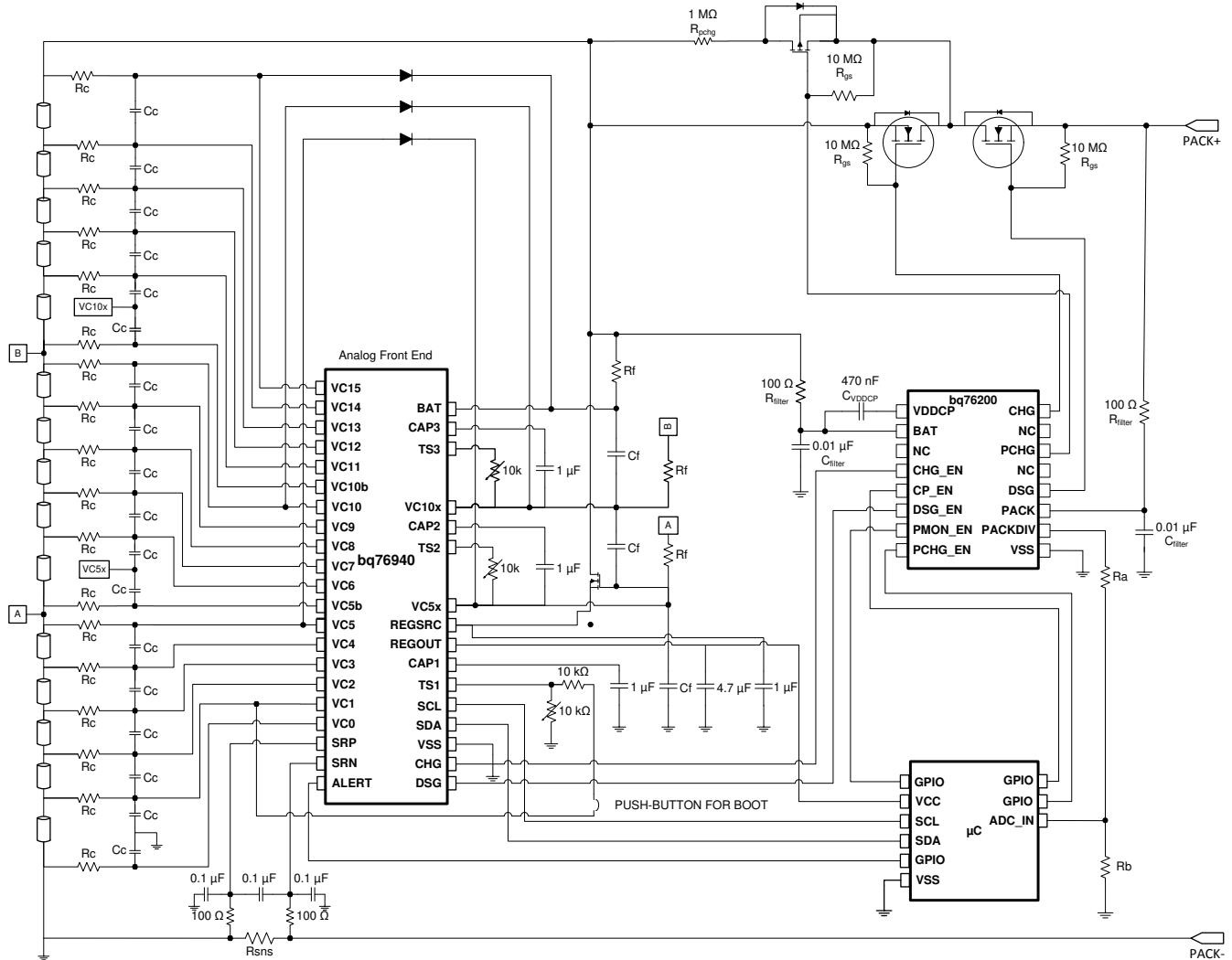


图 9-4. BQ76940 with BQ78350 Companion Controller IC and BQ76200 High-Side N-Channel FET Driver

## 9.2.1 Design Requirements

表 9-5. BQ769x0 Design Requirements

| DESIGN PARAMETER                 | EXAMPLE VALUE at T <sub>A</sub> = 25°C |
|----------------------------------|----------------------------------------|
| Minimum system operating voltage | 24 V                                   |
| Cell minimum operating voltage   | 3.0 V                                  |
| Series Cell Count                | 8                                      |
| Charge Voltage                   | 33.6 V                                 |
| Maximum Charge Current           | 3.0 A                                  |
| Peak Discharge Current           | 10.0 A                                 |
| OV Protection Threshold          | 4.30 V                                 |
| OV Protection Delay              | 2s                                     |
| UV Protection Threshold          | 2.5 V                                  |
| UV Protection Delay              | 4s                                     |
| OCD Protection Threshold Max     | 15 A                                   |
| OCD Protection Delay Time        | 320 ms                                 |
| SCD Protection Threshold Max     | 25 A                                   |